

RESEARCH ARTICLE

General Obstetrics

A case-control study of the interaction of fetal heart rate abnormalities, fetal growth restriction, meconium in the amniotic fluid and tachysystole, in relation to the outcome of labour

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Abstract

Objective: To quantify the incidence of intrapartum risk factors in labours with an adverse outcome, and compare them with the incidence of the same indicators in a series of consecutive labours without adverse outcome.

Design: Case-control study.

Setting: Twenty-six maternity units in the UK.

Population or sample: Sixty-nine labours with an adverse outcome and 198 labours without adverse outcome.

Methods: Observational study.

Main outcome measures: Incidence of risk factors in hourly assessments for 7 hours before birth in the two groups.

Results: A risk score combining suspected fetal growth restriction, tachysystole, meconium in the amniotic fluid and fetal heart rate abnormalities (baseline rate and variability, presence of decelerations) gave the best indication of likely outcome group.

Conclusions: Accurate risk assessment in labour requires fetal heart rate abnormalities to be considered in context with additional intrapartum risk factors.

KEY WORDS

fetal compromise, labour outcome, risk factors

1 | INTRODUCTION

A diagnosis of 'fetal distress' in labour has been commonly used for the last century, even though this is too broad to distinguish between the different challenges the fetus faces during labour.¹ Since the 1960s, continuous electronic recording and assessment of the fetal heart rate and uterine contractions (cardiotocography [CTG]) during labour have been widely adopted in high-income countries, and although a normal pattern is highly predictive of a normal outcome,² pattern abnormalities are suboptimal predictors of poor outcomes with high false-positive rates³ and

therefore do not enable us to determine accurately from CTG alone which baby would benefit from an earlier intervention. There have been attempts to systematise the classification of fetal heart rate patterns (e.g. in guidelines from the American College of Obstetricians and Gynecologists,⁴ the National Institute for Health and Clinical Excellence,⁵ the International Federation of Gynecology and Obstetrics⁶), as well as a model of interpretation, so-called 'Physiological interpretation',⁷ but they have variable sensitivity and specificity for adverse fetal outcomes⁸ and a consensus has been difficult to achieve. The national guidelines recommend risk factors that are associated with poor outcomes in addition to

the fetal heart rate, but they are not integrated into clinical decision-making.

Between 2015 and 2018, the Royal College of Obstetricians and Gynaecologists organised a national UK audit of intrapartum stillbirths, neonatal deaths, and babies admitted to a neonatal intensive care unit with evidence of hypoxic brain injury – *Each Baby Counts*.⁹ The incidence of substandard intrapartum care identified in labours with these adverse outcomes ranged between 71% and 76%, and in the 2015 report⁹ there was an average of six clinical risk factors associated with adverse outcome. The report authors recommended that consideration of the following risk factors, in addition to the fetal heart rate, should be standard practice for intrapartum fetal surveillance: reduced fetal movements, fetal growth restriction, previous caesarean section, meconium, suspected infection (pyrexia), vaginal bleeding and prolonged labour.

Between 10 January 2010 and 31 August 2013, 47 062 women in labour in 24 UK maternity units were randomly assigned to either a computerised system for interpreting CTGs, or analysis by the attending clinicians – the ‘INFANT’ trial (46 042 were available for analysis).¹⁰ There was no significant difference in the incidence of intrapartum death or poor neonatal outcome between the groups. A subsequent study reported that the risk factors associated with the 71 cases of poor outcome were body mass index of 30 kg/m² or more, hypertension, previous caesarean section, suspected fetal growth restriction, in vitro fertilisation conception, induction of labour with prostaglandins, oxytocin administration, meconium in the amniotic fluid, labour longer than 8 hours and pyrexia (temperature $\geq 37.5^{\circ}\text{C}$).

The hypothesis investigated in the current study was that considering and integrating risk factors in addition to abnormal fetal heart rates would improve intrapartum discrimination between labours with an abnormal versus a normal outcome.

2 | METHODS

On 15 August 2019 a joint committee was set up between the Royal College of Obstetricians and Gynaecologists and the Royal College of Midwives to establish a consensus approach to intrapartum fetal monitoring. Following a consensus workshop on 3 March 2020 where more than 90 attendees, including policy-makers, stakeholders, shop floor clinicians and women, gave their views on the concept, a working group was set up to assess whether a new approach aimed at (1) simplifying CTG classification and (2) incorporating additional risk factors, was feasible. This group met on 16 October 2020, to agree the fetal heart rate variables and risk factors to be included in an assessment of hourly intrapartum risk evaluation (Figure S1). There was no additional patient or public involvement in the design and planning of this study.

The initial analysis we report here is a comparison of the 71 cases of adverse outcome in the INFANT

trial – intrapartum stillbirth ($n = 3$), neonatal death ($n = 10$) or admission to neonatal intensive care unit with a metabolic acidosis ($n = 58$) – collected between 6 January 2010, and 31 August 2013 and a series of 199 consecutive births collected at four NHS maternity units in England (50 each in three units and 49 in one) between 1 January and 31 May 2021 and chosen as a non-purposive convenience sample.

Research ethics committee approval for the INFANT trial was granted by the National Research Ethics Service – Northern and Yorkshire Research Ethics Committee (09/H0903/31). Permission to further analyse and report data from the trial was gained from the original lead researcher (Prof. P. Brocklehurst) and the four members of the 2019 adverse outcomes appraisal group. Permission from the individual women in the study could not be obtained because the data were anonymised. A detailed analysis of their labours has already been performed and reported in 2019.¹¹ The labours from a maximum of 7 hours before birth were reassessed by PJS from an electronic copy of the CTG and the original notes made by the 2019 appraisal group and coded into the form shown in Figure S1 (CTG monitored in all cases). Two cases could not be coded and were removed from the analysis. In one case, an initial CTG when induction of labour was commenced was normal and the CTG was discontinued overnight; when it was recommenced 15 hours later the fetus had died. In another, the electronic copy of the CTG was corrupted and unreadable and the notes made by the 2019 appraisal group were insufficiently detailed for coding.

A comparison group was recruited from four maternity units in England during January to May 2021. In each unit, on a convenient day, 50 consecutive labours were audited retrospectively by MN, ZU, MB and KL and coded using the form illustrated in Figure S1. The audit committee in each unit gave permission to use the data for research. The data items captured are listed in Table 1 and were captured by inspection of the cardiotocogram and clinical records including the partogram. Data were anonymised and password protected before being centralised and analysed by PJS and CY. In one centre, the auditor miscounted and only 49 cases were audited. In another case, the data recorded were missing the analysis for the last 2 hours of labour, and this case was excluded. This left 198 cases for analysis. The labours were coded by the local auditor using the forms illustrated with an example in Figure S1, and then checked by CY when the results were transferred to a single data set for analysis. Some of the labours were longer than 7 hours, but there were too few for us to draw meaningful conclusions about the significance of scores in early labour and we therefore chose 7 hours as a practical cutoff for analysis.

Coding the risk factors was not strictly defined; the assessment was left to the clinical judgement of the auditor because this was thought to be the most clinically practical and generalisable approach. Fetal growth restriction (FGR) was considered as a risk factor if the coder identified expressions of concern within the clinical record regarding suspected inadequate growth of the fetus. We derived the risk factors

TABLE 1 Risk scores and presence of risk factors during labour by INFANT group versus comparative births group

Variable	INFANT, <i>n</i> = 69	Audit, <i>n</i> = 198	<i>p</i>
Delay in labour progress (<1 cm cervical dilatation in 2 h)	16 (23.2)	47 (23.7)	1.0
Oxytocin augmentation	43 (62.3)	76 (38.4)	0.001
Tachysystole (contractions ≥ 5 in 10 min)	34 (49.3)	33 (16.7)	<0.001
Fetal growth restriction suspected	10 (14.5)	9 (4.5)	0.012
New or increased meconium-stained liquor	18 (26.1)	26 (13.1)	0.022
Pyrexia (37.5°C $\times 2$ or 38.0°C $\times 1$)	8 (11.6)	16 (8.1)	0.463
Intrapartum bleeding per vaginam	4 (5.8)	5 (2.5)	0.243
Rise in fetal heart rate >20 bpm (IA only)		2/48 (4.3)	
CTG monitored cases only	<i>N</i> = 69	<i>N</i> = 160	
Abnormal baseline (≤ 100 bpm or ≥ 160 bpm)	36 (52.2)	13 (8.1)	<0.001
Rise in baseline >20 bpm	19 (17.5)	35 (21.9)	0.397
Abnormally reduced variability (absent or <5 bpm for >30 min)	43 (62.3)	14 (8.8)	<0.001
Repeated decelerations (any type)	61 (88.4)	76 (47.5)	<0.001
Prolonged decelerations (>3 min)	49 (71)	25 (15.6)	<0.001
Delivery by caesarean section	40/69 (58)	39/198 (19.7)	<0.001

Note: Columns show numbers of cases with the risk factor (%) identified at least once during labour.

from the *Each Baby Counts* studies and did not use a core outcome set.

2.1 | Statistical analysis

The data were collected using EXCEL spreadsheets and analysed using SPSS version 26 (IBM, Armonk, NY, USA). We analysed the frequency and percentage incidence of the risk factors described above. We quantified the relationship of the risk factors in relation to group membership (INFANT group versus comparative births group) using logistic regression.

3 | RESULTS

All the labours in the INFANT trial were monitored by CTG. In the comparative births group, the percentages monitored by CTG were 82.0%, 78.7%, 61.5% and 93.9% in the four units, respectively; the remainder were monitored with intermittent auscultation. The incidence of risk factors in the two cohorts is shown in Table 1. In the INFANT (adverse outcome) group, the most common risk factors were oxytocin augmentation (62.3%) and tachysystole (49.3%), and repeated (88.4%) or prolonged (71%) decelerations (not mutually exclusive). Abnormal (reduced) baseline variability (62.3%) and rate (mainly tachycardia, 52.2%) were also common. In the comparison (normal outcome) group oxytocin augmentation was significantly less common at 38.4% and tachysystole was significantly and substantially less frequent (16.7%). Abnormal (reduced) baseline variability (8.8%) and rate (mainly tachycardia, 8.1%) were also significantly less common. New or increased meconium-stained liquor was

TABLE 2 Risk factors predicting membership of the INFANT group versus comparative births group, using logistic regression

Risk factor	B	SE	Odds ratio	Significance
Fetal growth restriction	2.680	0.864	14.6	0.002
Meconium	1.502	0.596	4.5	0.012
Tachysystole	1.117	0.530	3.1	0.035
Abnormal baseline	1.993	0.537	7.3	<0.001
Reduced variability	2.412	0.529	11.2	<0.001
Decelerations	1.314	0.619	3.7	0.034
Prolonged decelerations	2.462	0.514	11.7	<0.001

Abbreviations: B, intercept; SE, standard error of B, significance is two-tailed.

twice as common in the INFANT group (26.1%) as in the comparison group (13.1%).

In the cases where CTG had been used, the risk factors were tested for their ability to predict which group (INFANT or comparative births group) each case belonged to, using logistic regression. This correctly predicted 52/69 (75.4%) cases as belonging to the INFANT group, and 151/160 (94.4%) cases as belonging to the comparative births group without adverse outcomes. The relative importance of the variables remaining significant in logistic regression is shown in Table 2. Suspected FGR was associated with the highest odds of discrimination (odds ratio 14.6), but it was relatively uncommon, being present in only 14.5% of the INFANT group and 4.5% of the comparative births group. With respect to the fetal heart rate, abnormal variability (odds ratio 11.2) and prolonged decelerations (odds ratio 11.7) were the best discriminators.

To assess whether there was a possible threshold for risk categorisation, we added up the number of risk factors

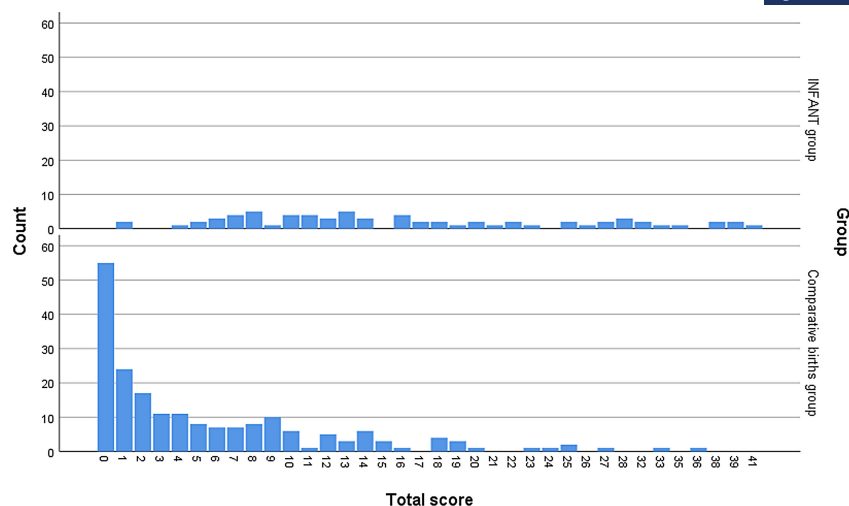


FIGURE 1 Total risk score comparing the INFANT group with the comparative births group

present at each hour of assessment. This resulted in a score that represented the number of risk factors present, with no attempt to weight the individual significance of each risk factor separately (weighting of the risk factors is being studied in a cohort of over 300 000 births in a single geographical area and will be reported separately). An hourly score of 4 provided the best discrimination, with 51/69 (73.9%) of the INFANT group reaching a score of 4 at least 1 hour before birth, compared with only 22/198 (11.1%) of the consecutive births group ($p < 0.001$).

To account for the duration of risk, we added the hourly totals to produce a total score for the labour up to 7 hours before birth. A persistent risk factor such as suspected FGR would add a number to the total score equivalent to the duration of labour in hours, whereas an intermittent risk factor such as a final prolonged deceleration would only add one per hour. The distribution of total scores by group is shown in Figure 1. The best discrimination was obtained using a total score cutoff of 10. Of the INFANT group, 47/69 (68.1%) had a total score of 10 or more compared with only 34/198 (17.2%) of the comparative births group. The receiver operating characteristic of the total score is shown in Figure 2. The area under the curve is 0.851 (95% confidence intervals 0.804–0.898).

4 | DISCUSSION

4.1 | Main findings

Our findings support the hypothesis that clinical risk factors improve the discrimination between a cohort with a known adverse outcome and a sequential sample of births in four English maternity units with normal outcomes, compared with that provided by fetal heart rate variables alone. Suspected fetal growth restriction was associated with the highest odds of discrimination (odds ratio 14.6), but it was relatively uncommon, being present in only 14.5%

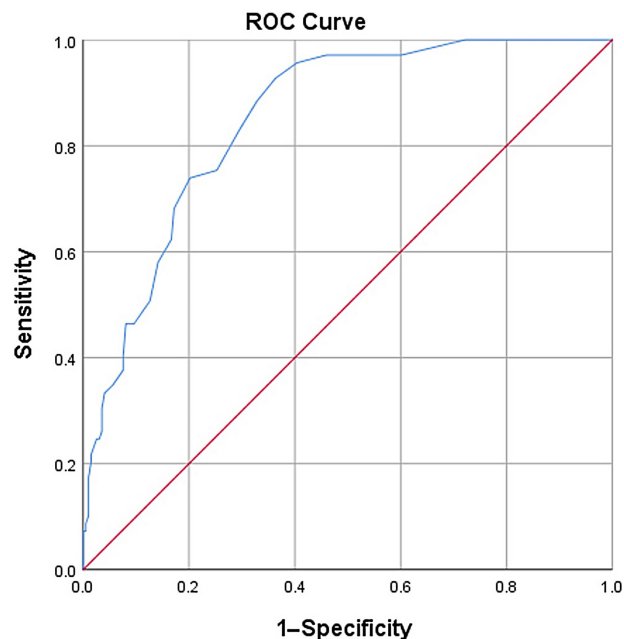


FIGURE 2 Receiver operating curve of total score in the prediction of group membership

of the INFANT group and 4.5% of the consecutive births group. With respect to the fetal heart rate, abnormal variability (odds ratio 11.2) and prolonged decelerations (odds ratio 11.7) were the best discriminators. Meconium staining of the amniotic fluid (odds ratio 4.5) was much more common in the INFANT group than in the comparative births group (26.1% versus 13.1%) and tachysystole (odds ratio 3.1) was even more common (49.3% versus 16.7%). When tachysystole was taken into account, slow progress and the administration of oxytocin, although common, were no longer significant discriminators, suggesting that it is not the use of oxytocin in slow labour that is a problem, but its misuse.

Khalil et al.¹² reviewed intrapartum risk factors and also reported that meconium-stained liquor, oxytocin use, intrapartum haemorrhage, and pyrexia were significant factors associated with operative delivery for fetal compromise.

A more complex approach has been proposed by Eden et al.¹³ who advocate the use of a 'fetal risk index' that includes baseline fetal heart rate, variability and decelerations, and 'abnormal uterine activity' (more complex than tachysystole). However, they also include assessment of ten maternal risk factors, nine obstetrical risk factors, and twelve fetal risk factors. Although these variables are comprehensive, they are complex to assess, and it is not clear from the published data how they can be incorporated into real-time clinical practice.

4.2 | Strengths and limitations

A strength was the use of a well-characterised and previously reported cohort as the group with an adverse outcome. Another strength was the use of consecutive births at four separate maternity units, which helps to reduce bias from selective recruitment of the control group. While relying on the individual opinion of the auditors to assign risk rather than using multiple assessors and closely defined criteria might be seen as a weakness, we considered that this would be more representative of usual clinical practice, and therefore more generalisable.

4.3 | Interpretation

There is a recognised association between FGR and the ability of the fetus to withstand the stresses of labour and it should therefore be included in decision-making when interpreting fetal heart rate abnormalities in labour. However, there is no 'cliff-edge' change in risk, instead, the smaller the baby, the greater the incidence of a poor outcome.¹⁴ Hence, any suspicion of smallness should lower the threshold for concern, even if the pregnancy is otherwise low risk.¹⁵

Although meconium in the amniotic fluid is increasingly common after 37 weeks of gestation,¹⁶ and is relatively innocuous if the fetal heart rate pattern is normal,¹⁷ there is an increase in the risk of a poor outcome if the contemporaneous fetal heart rate tracing is even moderately abnormal.¹⁸ Both slow labour and oxytocin infusion are common in labours with a poor outcome; however, it is reassuring that they no longer appear significant once tachysystole is taken into account. This suggests that careful attention to contraction frequency and reducing/stopping oxytocin infusions immediately if there is tachysystole will safeguard the fetus.

Our approach to integrating a small set of important clinical risk factors with the fetal heart rate assessment is simplistic, and it is likely that their interactions are not simply additive or linear. Therefore, we are currently undertaking further work on this aspect that will focus on developing a model that can deal with these complexities, while still

being easy to use in routine clinical practice. In brief, the risk factor approach we have described has been adapted into new documentation similar to a MEWS (Maternity Early Warning Score) chart which also incorporates elements of the partogram.

Finally, an accurate risk assessment in labour where fetal heart rate abnormalities are considered in context with the presence of additional intrapartum risk factors will facilitate personalisation in intrapartum care. Our study has been a component in the development of a new national initiative in the UK called 'ABC' (Avoiding Brain Injury in Childbirth). This is a collaboration between the Royal College of Obstetricians and Gynaecologists, the Royal College of Midwives, and The Health Improvement Studies Institute (THIS, Cambridge), with funding from the UK Department of Health and Social Care.

5 | CONCLUSION

Clinical decision-making for fetal monitoring in labour should not focus entirely on the fetal heart rate pattern but should be integrated with other important clinical risk factors, notably FGR, meconium in the amniotic fluid and tachysystole.

AUTHOR CONTRIBUTIONS

The study was conceived by PJS, TD, CW and MB. Data collection in the comparative births group was by MB, KL, MN and ZU. Data collation was by CW, CWHY and PJS. Data analysis was by PJS and CWHY. The first version of the paper was written by PJS and CWHY. All authors commented and provided feedback on and approved the final version for publication.

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The study was not specifically funded.

CONFLICT OF INTERESTS

PJS is the medical director and a shareholder in CaretekMedical, a company that has produced a smartphone app for women to assess their antenatal risk factors (<http://www.caretekmedical.net/>). Details of his additional interests can be found at https://obgyn.onlinelibrary.wiley.com/hub/journal/14710528/editors-disclosures-of-interests#Philip_Steer. CWHY was seconded to the Royal College of Obstetricians & Gynaecologists/UK Department of Health and Social Care for this work. He is a faculty member of the PROMPT Maternity Foundation, a charity involved in researching and delivering multi-professional training for obstetric emergencies and has previously been partially funded by the PROMPT Maternity Foundation. MB, KL, MN and

ZU declare that there are no conflicts of interest. CW is the lead Midwife for the PROMPT Maternity Foundation charity (www.promptmaternity.org). TD is the Vice-President (clinical quality) of the Royal College of Obstetricians and Gynaecologists. Completed disclosure of interests form available to view online as supporting information.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

ETHICS APPROVAL

Research ethics committee approval for the INFANT trial was granted by the National Research Ethics Service – Northern and Yorkshire Research Ethics Committee (09/H0903/31). Permission to further analyse and report data from the trial was gained from the original lead researcher (Prof. P Brocklehurst) and the four members of the 2019 adverse outcomes appraisal group. Permission from the individual women in the study could not be obtained because the data were anonymised. With regard to the comparative births group, the audit committee in each unit gave permission to use data collected routinely for clinical purposes to be analysed for research. In the United Kingdom, the Health and Social Care Act of 2011 established that ethics committee approval for routinely collected data to be used by the clinical team for audit and quality assurance was not required. Data were anonymised and password protected before being centralised.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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